INCH - POUND

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MILITARY SPECIFICATION

INSULATING COMPOUND, ELECTRICAL, EPOXY, COLLOIDAL SILICA FILLED FOR POTTING AND ENCAPSULATION

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an epoxy resin electrical insulating compound used for potting and encapsulating.

1.2 Classification. The epoxy resin shall be of the following types, as specified (see 6.1 and 6.2):

Type I	-	Low viscosity
Type II	-	Medium viscosity
Type III	-	High viscosity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document tot he extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 257	-	D-C Resistance or Conductance of Insulating Materials, Standard Test Methods for
ASTM D 1084	-	Viscosity of Adhesives, Standard Test Methods for
ASTM D 2240	-	Rubber Property-Durometer Hardness, Standard Test Method for
ASTM D 3951	-	Commercial Packaging, Standard Practice for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials.

3.2.1 Composition and condition in container (see 4.7.4.1).

3.2.1.1 Composition. The materials furnished under this specification shall be premixed, frozen, degassed compounds formulated from the ingredients specified in table I.

TABLE I. Material composition.

Ingredients	Parts by weight		
	Type I	Type II	Type III
Epoxy resin I	50	50	50
Epoxy resin II	50	50	50
Amine curing agent	19.5	19.5	19.5
Colloidal silica	2.5	3.5	5.0

3.2.1.2 Condition in container. The compounds, as received, shall be free of gel particles, dirt, or any other contaminants which would affect the intended use.

3.2.2 Pot life. The useable pot life of the compounds shall be not less than 30 minutes (see 4.7.4.2).

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3.2.3 Viscosity. The viscosity of the thawed materials, as received, shall be as follows (see 4.7.4.3):

Type I	300 ± 100 centipoises (cP)
Type II	850 ± 200 cP
Type III	7500 ± 2000 cP

3.2.4 Storage life. The premix, as furnished by the manufacturer, shall be capable of meeting the requirements specified herein after storage at a temperature of less than 40°C in an original closed container for not less than 3 months (see 4.7.4.4).

3.3 Performance characteristics.

3.3.1 Porosity. The cured compound shall exhibit no bubbles or porosity smaller than 0.031 inch diameter (see 4.7.4.5).

3.3.2 Hardness. The cured compound shall have a hardness value of not less than 40 Shore D as specified by ASTM D 2240 (see 4.7.4.6).

3.3.3 Flexibility. Bending shall cause no chipping or cracking (see 4.7.4.7).

3.3.4 Insulation resistance. The insulation resistance shall be not less than 1×10^{11} ohms (see 4.7.4.8).

3.4 Workmanship. The compound shall be uniform in quality and shall be free of dark colored contaminants, lumps, foreign materials, and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspections system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the

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responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. As required in paragraph 3.1 and unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein (see 4.6).

4.4 First article inspection. When specified in the contract or order (see 6.3), a first article sample consisting of 8 ounces of each type of the frozen premix in sealed tubes or cans, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to first article inspection. Accomplishment of first article inspection shall be as specified herein. If the first article sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the first article sample. Units subjected to first article inspection shall have successfully passed acceptance inspection. Testing of the first article sample to determine compliance shall be in accordance with the corresponding test paragraphs.

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4.5 First article inspection. First article inspection shall be as specified in table II.

TABLE II. First article inspection.

Characteristic	Requirement paragraph	Test paragraph
Composition and condition in container	3.2.1	4.7.4.1
Pot life	3.2.2	4.7.4.2
Viscosity	3.2.3	4.7.4.3
Storage Life	3.2.4	4.7.4.4
Porosity	3.3.1	4.7.4.5
Hardness	3.3.2	4.7.4.6
Flexibility	3.3.3	4.7.4.7
Insulation resistance	3.3.4	4.7.4.8
Workmanship	3.4	4.9

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4.5.1 Sampling. Unless otherwise specified in the contract or order (see 6.2), sampling shall consist of a random sample selected from each production lot in accordance with table III and will be tested as specified in paragraph 4.5.3.

TABLE III. Sampling plan for quality conformance.

Lot size	Sample size
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 to 150,000	40
150,001 to 500,000	40
500,001 and over	40
Accept on Zero and reject on 1 or more.	

4.5.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.5.3 Quality conformance inspection. Quality conformance inspection of the sample specified in 4.5.1, to determine compliance with the characteristics specified in Table IV, shall be conducted in accordance with the corresponding tests and inspection paragraphs.

TABLE IV. Quality conformance inspection.

Characteristic	Requirement paragraph	Test paragraph
Composition and condition in container	3.2.1	4.7.4.1
Porosity	3.3.1	4.7.4.5
Hardness	3.3.2	4.7.4.6
Workmanship	3.4	4.9

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4.6 Inspection equipment. The inspection equipment for conducting tests shall be as specified in the applicable test methods and procedures paragraphs.

4.7 Test methods and procedures.

4.7.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

(a)	Temperature	Room ambient ± 9°C
(b)	Altitude	Facility ground
(c)	Humidity	Facility ambient up to 95 percent relative humidity

4.7.2 Test sequence. Testing sequence within each classification of inspection (first article or acceptance) shall be at the option of the supplier.

4.7.3 Test specimen preparation. The frozen premix compound shall be allowed to thaw at room temperature of not less than 18°C and not greater than 38°C, prior to application. The compound, after application or in its original container as specified, shall be cured with one of the following cure cycles:

(a) Not less than 48 hours at 27 ± 3°C

(b) Not less than 2 hours at 74 ± 3°C

4.7.4 Tests.

4.7.4.1 Composition and condition in container. The composition shall be certified by the contractor and the condition in container shall be examined visually to establish compliance to 3.2.1.

4.7.4.2 Pot life. A syringe of material shall be allowed to thaw to 24 ± 3°C. The material shall remain uncured and injectable for the pot life specified in 3.2.2.

4.7.4.3 Viscosity. The material shall be allowed to thaw to 24 ± 3°C. The viscosity shall then be determined in not less than five minutes or greater than 10 minutes after reaching the test temperature in accordance with ASTM D 1084 with the following exception: an LVF Brookfield viscometer or equal shall be used, operating as specified below.

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	Spindle number	Speed (revolutions per minute (rpm))
Type I	2	6
Type II	2	6
Type III	3	6

The viscosity shall conform to 3.2.3.

4.7.4.4 Storage life. A sufficient amount of premix in an original unopened container shall be stored under the conditions specified in 3.2.4. At the end of the storage period, the premix shall be subjected to all the test requirements specified herein. The contractor's certification of compliance certifying to the storage life requirements (see 3.2.4) may be accepted during the storage period.

4.7.4.5 Porosity. A cured sample shall be cut or broken and then examined under 10 power magnification to determine conformance to the requirement of 3.3.1.

4.7.4.6 Hardness. Hardness shall be determined on a cured specimen at ambient temperature in accordance with ASTM D 2240 and shall conform to the requirement of 3.3.2.

4.7.4.7 Flexibility. Two strips of cured material 0.125 inch thick, 0.25 inch wide, and 3.0 inches long shall be bent over a 2-inch diameter mandrel and visually examined for conformance to the requirement of 3.3.3. Retest, if necessary, shall be conducted in accordance with paragraph 4.7.4.9.

4.7.4.8 Insulation resistance. Testing shall be at ambient temperature in accordance with ASTM D 257, and the insulation resistance shall conform to the requirement in 3.3.4.

4.7.4.9 Retest. An acceptance sample of material which has passed the acceptance tests need not be retested until its shelf life has expired. At such time one container shall be selected at random from the overage acceptance sample. The material shall be thawed and examined for evidence or partial cure by observing its ability to be extruded. If material passes this phase, pot life test shall be conducted in accordance with 4.7.4.2 and samples for cure shall be prepared simultaneously. If samples cure satisfactorily, they shall be tested for hardness in accordance with 4.7.4.6 and flexibility in accordance with 4.7.4.7. Material which successfully passes retest shall be updated for one month from completion date of tests. Material shall not subsequently be retested, updated, or used.

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4.8 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.9 Workmanship. The cured compound shall be visually examined for conformance to the workmanship requirement specified in 3.4.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level C in accordance with MIL-STD-1190. Additional marking shall include, but not be limited to the following:

a. Lot or batch number

b. Date of manufacture

c. Type

d. Size of container

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. When specified in the contract (see 6.2) this compound is intended for use in potting and encapsulation of low voltage (less than 300 volts) cable connectors at temperatures from 62 to 71°C. The material is supplied in three compositions of differing viscosities.

Type I	-	Low viscosity material for use where connector pins have been previously sealed to prevent leak-through along the connector pins. Normally used when heat curing is desired.
Type II	-	Medium viscosity material used with or without pin sealing. If pin sealing is not performed with type III material, the type II material should be cured at room temperature.

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Type III	-	High viscosity material for use on connectors with large open areas or widely spaced pins, where the material has poor flow characteristics.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Type of compound (see 1.2 and 6.1)

d. Whether a first article is required (see 3.1)

e. Sampling plan for quality conformance (see 4.5.1)

f. Requirement for certification of composition (see 4.7.1)

g. Inspection of packaging (see 4.8)

h. Size of container

i. Quantity

j. Intended use (see 6.1)

k. Part identification number (see 6.5)

6.3 First article. When first article inspection is required (see 3.1), the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Composition. Compounds which meet the requirements of this specification may be made from ingredients with the characteristics specified in table V.

TABLE V. Composition.

Characteristic	Epoxy resin I	Epoxy resin II	Amine Curing agent	Filler
Viscosity, 25°C, cP	55-100	10,000-16,000	-	-
Epoxide equivalent	305-335	175-210	-	-
Amine Value	-	-	63-67	-
Silicon dioxide after removal of volatiles, weight, percent	-	-	-	99.00
Specific surface area, ft^2	-	-	-	16.26-20.90
Tap density, 25°C lbs/cu. ft	-	-	-	2.5-3.5

6.4.1 Description of compound ingredients. The compound ingredients are as follows:

(a) Epoxy resin I - A liquid epoxy, polyglycol, flexible when cured.

(b) Epoxy resin II - A liquid epoxy, bisphenol A, rigid when cured.

(c) Amine curing agent - A liquid aliphatic triamine.

(d) Filler - Colloidal silica.

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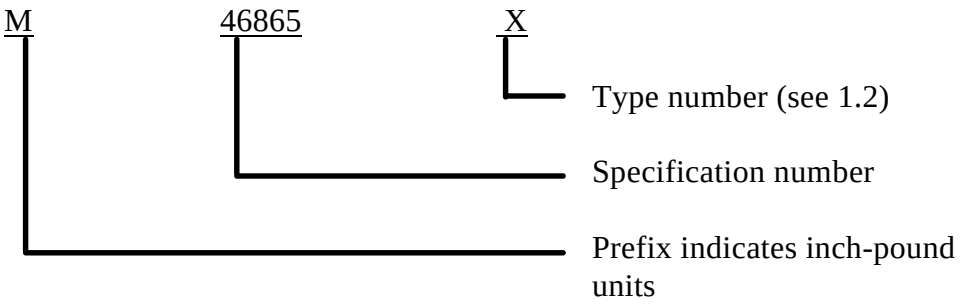
6.5 Part or identifying number.



6.6 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.7 Subject term (keyword) listing.

Hardener
Resin laminate
Premix
Crystalline substance

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
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